

PROGRAM 1 :

Q)

Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

CODE :

```
import csv

a = []
with open('enjoysport.csv', 'r') as csvfile:
    for row in csv.reader(csvfile):
        a.append(row)
print(a)
print("\nThe total number of training instances are:", len(a))
num_attribute = len(a[0]) - 1
print("\nThe initial hypothesis is:")
hypothesis = ['0'] * num_attribute
print(hypothesis)
for i in range(0, len(a)):
    if a[i][num_attribute] == 'Yes':
        for j in range(0, num_attribute):
            if hypothesis[j] == '0' or hypothesis[j] == a[i][j]:
                hypothesis[j] = a[i][j]
            else:
                hypothesis[j] = '?'
    print("\nThe hypothesis for training instance {} is:\n".format(i + 1), hypothesis)
print("\nThe Maximally Specific Hypothesis is:")
print(hypothesis)
```

OUT PUT :

```
[[['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'EnjoySport'], ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'], ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'], ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'], ['Same', 'Same', 'Same', 'Same', 'Same', 'Same', 'Same']]]

The total number of training instances are: 5

The initial hypothesis is:
['?', '?', '?', '?', '?', '?']

The hypothesis for training instance 1 is:
['?', '?', '?', '?', '?', '?']

The hypothesis for training instance 2 is:
['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

The hypothesis for training instance 3 is:
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

The hypothesis for training instance 4 is:
['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

The hypothesis for training instance 5 is:
['Sunny', 'Warm', '?', 'Strong', '?', '?']

The Maximally Specific Hypothesis is:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```